methods

1.1 Sample Quality Control

Please refer to Novogene's QC report for methods of sample quality control.

1.2 Library Construction, Quality Control and Sequencing

Sampling 1 μg of genomic DNA, the sample is randomly fragmented into segments of approximately 350 bp using a Covaris ultrasonic disruptor to construct the library. The entire library preparation is completed through steps including end repair, addition of A-tails, ligation of sequencing adapters, purification, and PCR amplification. After library construction, the integrity of the library fragments and the size of the inserted fragments are assessed using AATI analysis. If the insert size meets expectations, the accurate concentration of the effective library is quantified using Q-PCR (effective library concentration > 3 nM) to ensure the library quality. After the library passes the quality check, different libraries are pooled according to their effective concentrations and target data output requirements, and then subjected to PE150 sequencing.

2 Bioinformatics Analysis Pipeline

2.1 Preprocessing of sequencing results

Fastp (https://github.com/OpenGene/fastp) is used for preprocessing raw data from the sequencing platform to obtain clean data for subsequent analysis. We will discard the paired reads in the following situation: when either one read contains adapter contamination; when either one read contains more than 10 percent uncertain nucleotides; when either one read contains more than 50 percent low quality nucleotides (base quality less than 5).

Considering the possibility of host contamination in samples, clean data needs to be blasted to the host database to filter out reads that may come from host origin. Bowtie2 software (http://bowtie-bio.sourceforge.net/bowtie2/index.shtml) is used by default, with the following parameter settings: --end-to-end, --sensitive, -I 200, and -X 400 (Karlsson FH et al., 2012; Karlsson FH et al., 2013; Scher JU et al., 2013).

2.2 Assembly of Metagenome

MEGAHIT software is used for assembly analysis of clean data, with assembly parameter settings:

--presets meta-large (--end-to-end, --sensitive, -I 200, -X 400) (Karlsson FH et al., 2013; Nielsen HB et

al., 2014), and scaftigs without N is obtained by breaking the resulted scaffolds from the N junction

(Qin J et al., 2010; Li D et al., 2015).

2.3 Binning

The Binning module uses MetaWRAP to integrate three mainstream metagenomic binning software: MaxBin2, MetaBAT2, and CONCOCT (Uritskiy et al., 2018). First, bwa-index is used to build an index for the metagenomic assembly results, and then paired reads from any number of samples are aligned to the assembly. The alignment results are sorted and compressed using samtools, while collecting statistical data on library insert fragment sizes (including the mean insert size and standard deviation). Next, the jgi_summarize_bam_contig_depths function from MetaBAT2 is used to generate a contig abundance table, which is then converted into input formats suitable for each binning software. MaxBin2, MetaBAT2, and CONCOCT complete the binning of contigs under default settings, ultimately generating a bin folder containing formatted bin fasta files, facilitating subsequent extended analysis based on bins.

The Bin_refinement module employs a hybrid approach to integrate two or three bin sets obtained from different software (or the same software with different parameters) into an improved bin set (Uritskiy et al., 2018). First, binning_refiner is used to generate hybrid bins from each possible combination. If there are three bin sets: A, B, and C, then binning_refiner will generate the following hybrid sets: AB, BC, AC, and ABC. Then, CheckM is run to evaluate the completeness and contamination of bins in each bin set (3 original sets and 4 hybrid sets) (Parks et al., 2015). Subsequently, the bin sets are iteratively compared, and each pair of bin sets is merged into an improved bin set. To this end, identical bins in the two bin sets are identified through at least 80% genome length overlap, and the best bin is determined based on the one with the higher score. The scoring function is S = completeness - 5 × contamination. After all bin sets are merged into the final bin set, the deduplication function removes any duplicate contigs: if a contig exists in multiple bins, it will be removed from all bins except the one with the highest score. Then, CheckM is run again to evaluate the final bin set and generate a final report file containing the completeness, contamination, and other statistical data generated by CheckM for each bin. Additionally, completeness and contamination ranking plots are generated to assess the success of the Bin_refinement module and compare its output with the quality of the original bins.

2.4 Genome Quality Assessment

After optimization in the Bin_refinement module, to integrate the bins results from all samples and remove potential redundancy, dRep is used to perform clustering and dereplication on the optimized bins (Olm et al., 2017). This step first conducts an initial screening of whole-genome similarity using Mash, followed by selecting representative bins based on the ANI (Average Nucleotide Identity) threshold (default 95%) and CheckM quality scores (including completeness and contamination), ultimately generating a project-wide dereplicated MAGs collection. This ensures the uniqueness and high quality of the bins, avoiding redundant representations that could affect downstream analysis, and provides a reliable foundation for subsequent final quality assessment.

CheckM is a key tool in metagenomic analysis for evaluating genome quality (Binning results). It estimates genome completeness and contamination by identifying specific marker genes (Parks et al., 2015). Its core advantage lies in selecting lineage-specific marker gene sets based on the genome's position in the phylogenetic tree, making the evaluation results more accurate, especially for MAGs analysis in complex microbial communities.

GTDB-Tk (Genome Taxonomy Database Toolkit) is a tool based on the Genome Taxonomy Database (GTDB) for standardized species annotation of bacterial and archaeal genomes (Chaumeil et al., 2019). Its core advantage is the use of whole-genome phylogenetic analysis rather than traditional 16S rRNA gene sequences, providing a more objective and accurate classification framework, particularly suitable for species classification of genome drafts (MAGs) obtained from metagenomic binning.

2.5 Gene Prediction and Functional Annotation

Gene prediction is a key step in metagenomic analysis, aimed at identifying protein-coding genes (CDS) and non-coding RNAs (such as rRNA, tRNA, tmRNA) from assembled sequences (such as contigs or scaffolds) or binning results, providing foundational data for downstream functional annotation and metabolic analysis. Prokka is a fast prokaryotic genome annotation tool specifically designed for efficient prediction and annotation of bacterial or archaeal genomes (Seemann, 2014). Its core methods include: using Prodigal for ab initio open reading frame (ORF) prediction to identify CDS; integrating tools such as RNAmmer and Aragorn to detect non-coding RNAs like rRNA, tRNA, and tmRNA; and incorporating homology alignments (such as BLAST searches against Swiss-Prot or UniProt) to enhance annotation accuracy.

dbCAN (Carbohydrate-active enzyme ANnotation) is a web server and toolkit dedicated to automated annotation of carbohydrate-active enzymes (CAZymes) (Zheng et al., 2023). The main purpose of dbCAN is to efficiently identify, classify, and predict the functions of CAZyme genes in genomic or metagenomic sequences by integrating multiple computational methods, helping researchers reveal the roles of microorganisms in carbohydrate metabolism, particularly suitable for metagenomic analysis.

eggNOG-mapper (evolutionary genealogy of genes: Non-supervised Orthologous Groups - mapper) is a tool dedicated to rapid whole-genome functional annotation through orthology assignment (Cantalapiedra et al., 2021). It is based on pre-computed orthologous groups and phylogenies from the eggNOG database, transferring functional information from fine-grained orthologs, avoiding the low precision issues in traditional homology searches (such as BLAST) caused by functional divergence of paralogs.

CARD (Comprehensive Antibiotic Resistance Database) is a bioinformatics database focused on the rigorous curation and annotation of antibiotic resistance genes (ARGs), their products, and related phenotypes (Alcock et al., 2023). The core of CARD is the Antibiotic Resistance Ontology (ARO), an ontology framework for organizing resistance determinants, antibiotics, and related data.

BacMet2 (Antibacterial Biocide and Metal Resistance Genes Database) is a bioinformatics database focused on the curation and annotation of bacterial resistance genes to metals and antibacterial biocides (such as disinfectants and preservatives) (Pal et al., 2014). The database is based on in-depth review of scientific literature, providing manually curated content, and linking with related resources such as antibiotic resistance databases.

VFDB (Virulence Factor Database) is a bioinformatics database focused on the curation, classification, and annotation of bacterial virulence factors (VFs), providing virulence-related knowledge for various bacterial pathogens (Chen et al., 2005).

Additionally, to elucidate the distribution of mobile genetic elements (MGEs) in microbial communities and their potential role in the horizontal transfer of antibiotic resistance genes (ARGs) and virulence factors (VFs), this study performed specialized MGE annotation on dereplicated MAG sequences. The primary tool employed was geNomad, a deep neural network-based framework integrated with marker gene databases, enabling efficient and accurate simultaneous identification and classification of viruses (including prophages), plasmids, and composite MGEs, with high sensitivity and low false-positive rates in metagenomic assembled sequences (particularly MAGs) (Camargo et al., 2024). geNomad's end-to-end mode directly conducts gene prediction, classification, and functional annotation on input FASTA sequences, yielding outputs that include element types, confidence scores, hallmark gene counts, and details on potential conjugation transfer genes and resistance genes. To address geNomad's limited coverage of insertion sequences (IS) and transposons, MobileElementFinder was used as a complementary tool, which rapidly detects IS, composite transposons, unit transposons, and integrative conjugative elements (ICEs) via alignment to a dedicated database (MGEdb), providing precise positional, directional, and typological information (Johansson et al., 2021). This integrated strategy enables comprehensive evaluation of MGE abundance and functional cargo, offering key insights into microbial community evolutionary dynamics, resistance dissemination mechanisms, and ecological risks.

3 Reference

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